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Appendix D

PORTFOLIO THEORY

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D.1 MinVar Frontier Mathematics

In we showed that the weight vector $\mathbf{w}_{mv}$ that minimize portfolio variance, subject to the mean return and full investment constraints $\mathbf{w}' = \mu_P$ and $\mathbf{w}'\mathbf{1} = 1$, has the form

$$\mathbf{w}_{\text{mv}} = \lambda_1 \mathbf{C}^{-1} + \lambda_2 \mathbf{C}^{-1}\mathbf{1}. \tag{D.1}$$

Applying the mean return and full-investment constrains results in two equations in the two unknowns λ_1 and λ_2, and the reader can easily show that the solution to these equations is

$$\begin{aligned} \lambda_1 &= \frac{c\mu_P - a}{d} \\ \lambda_2 &= \frac{b - a\mu_P}{d} \end{aligned} \tag{D.2}$$

where $a = \mathbf{1}'\mathbf{C}^{-1}$, $b =' \mathbf{C}^{-1}$, $c = \mathbf{1}'\mathbf{C}^{-1}\mathbf{1} > 0$, and $d = bc - a^2$.

Substituting λ_1 and λ_2 into (D.1) and easy algebraic manipulation results in

$$\mathbf{w}_{\text{mv}} = \mathbf{w}_1 + \mu_P \mathbf{w}_2 \tag{D.3}$$

where

$$\begin{aligned} \mathbf{w}_1 &= \frac{1}{d}\left(b\mathbf{C}^{-1}\mathbf{1} - a\mathbf{C}^{-1}\right) \\ \mathbf{w}_2 &= \frac{1}{d}\left(c\mathbf{C}^{-1} - a\mathbf{C}^{-1}\mathbf{1}\right). \end{aligned} \tag{D.4}$$

and

$$d = (' \mathbf{C}^{-1})(\mathbf{1}'\mathbf{C}^{-1}\mathbf{1}) - (\mathbf{1}'\mathbf{C}^{-1})^2 \tag{D.5}$$

$$\geqslant 0 \tag{D.6}$$

where we show below how the above inequality follows from the Cauchy-Schwarz inequality, and that d is positive except in the special case where $= c\mathbf{1}$ for some constant c, which does not arise in practice.

Also, we note that

$$\begin{aligned} c^{-1} &= \sigma^2_{\text{gmv}} \\ \frac{a}{c} &= \mu_{\text{gmv}} \end{aligned} \tag{D.7}$$

where μ_{gmv} and σ^2_{gmv} are the mean and variance of the global minimum variance (GMV) portfolio.

Positivity of d

Since $\mathbf{C}$ is positive definite so is $\mathbf{C}^{-1}$, and basic matrix theory shows that therefore $\mathbf{C}^{-1}$ can be factored into the product of two inverse square root matrices, $\mathbf{C}^{-1} = \mathbf{C}^{-1/2}\mathbf{C}^{-1/2}$ where $\mathbf{C}^{-1/2}$ is positive definite. Thus one has

$$
\begin{aligned}
d &= ({}'\mathbf{C}^{-1/2}\mathbf{C}^{-1/2}) \cdot (\mathbf{1}'\mathbf{C}^{-1/2}\mathbf{C}^{-1/2}\mathbf{1}) - (\mathbf{1}'\mathbf{C}^{-1/2}\mathbf{C}^{-1/2}\boldsymbol{\mu})^2 \\
&= (\check{}) \cdot (\check{\mathbf{1}}'\check{\mathbf{1}}) - (\check{\mathbf{1}}^{\prime})^2
\end{aligned}
$$

where $\check{} = \mathbf{C}^{-1/2}$ and $\check{\mathbf{1}} = \mathbf{C}^{-1/2}\mathbf{1}$. The Cauchy-Schwarz inequality for two vectors $\mathbf{x}$ and $\mathbf{y}$ states that

$$
(\mathbf{x}'\mathbf{y})^2 \leqslant (\mathbf{x}'\mathbf{x}) \cdot (\mathbf{y}'\mathbf{y}) \quad \text{with equality if, and only if,} \quad \mathbf{y} = c \cdot \mathbf{x} \tag{D.8}
$$

for some non-zero constant c. Thus d is equal to zero if and, only if, $\mathbf{C}^{-1/2} = c \cdot \mathbf{C}^{-1/2}\mathbf{1}$, equivalently if and only if $\mathbf{C}^{-1/2}(-c \cdot \mathbf{1}) = \mathbf{0}$, where $\mathbf{0}$ is a vector of zeros. But since $\mathbf{C}^{-1/2}$ is positive definite, the condition $\mathbf{C}^{-1/2}(-c \cdot \mathbf{1}) = \mathbf{0}$ can only occur if $= c \cdot \mathbf{1}$. Thus $d = 0$ if, and only if, the portfolio mean vector has all equal values, which essentially never happens in practice.

D.2 The Tangent Portfolio is the Market Portfolio Under the CAPM

Under the first CAPM assumption in , all market participants hold the tangent portfolio T, and the last assumption implies that the supply of securities equals the demand. Let N be the number of securities in the market, with V_j be the market capitalization of the jth security. Then $V = \sum_{j=1}^{N} V_j$ is the total market capitalization, and $w_{M,j} = V_j/V$ is the weight of the jth security in the market portfolio. Let K be the number of market investors, where the kth investor invests an amount $\gamma_k W_k,\ 0 < \gamma_k < 1$ in risky assets and an amount $W_k(1-\gamma_k)$ in the risk-free asset. In general the kth investor would invest an amount $\gamma_k W_k w_{k,j}$ in the jth market asset, and therefore the total amount invested in the jth asset by all investors is

$$
\dot{V}_j = \sum_{k=1}^{K} \gamma_k W_k w_{k,j}.
$$

However, under the CAPM assumptions all investors all hold the same tangent portfolio, so $w_{k,j} = w_j, k = 1, 2, \cdots, K$ where w_j is the weight of the jth asset in the tangent portfolio T, and $\sum_{j=1}^{N} w_j = 1$. Thus

$$
\dot{V}_j = w_j \sum_{k=1}^{K} \gamma_k W_k
$$

is the amount invested by all investors in the jth asset, and $\dot{V} = \sum_{j=1}^{N} \dot{V}_j = \sum_{k=1}^{K} \gamma_k W_k$ is the total amount invested by all investors. However, under the CAPM equilibrium assumption $\dot{V}_j = V_j$ and $\dot{V} = V$, which results in

$$
w_{M,j} = \frac{V_j}{V} = \frac{\dot{V}_j}{\dot{V}} = w_j,\ j = 1, 2, \cdots, N
$$

and so the tangent portfolio T is the market portfolio M.

D.3 Proof of the CAPM

The following is a simple proof of the CAPM formula with beta defined by . Let M be the market portfolio whose mean return and variance are μ_M and σ_M^2 respectively. As the market portfolio is the tangent portfolio, the Capital Allocation Line associated with the entire market is known as the *Capital Market Line* (CML), and can be written as

$$\mu = r_f + \frac{(\mu_M - r_f)}{\sigma_M}\sigma. \tag{D.9}$$

Now consider any security in the market place with return r_i, mean μ_i and variance σ_i^2, such that (μ_i, σ_i) does not lie on the efficient frontier. We form a portfolio P by investing a fraction α in the security and a fraction $1-\alpha$, in the market portfolio M. The mean return and variance of the portfolio P are:

$$\begin{aligned}
\mu_P &= \alpha\mu_i + (1-\alpha)\mu_M \\
\sigma_P &= \left(\alpha^2\sigma_i^2 + 2\alpha(1-\alpha)\sigma_{i,M} + (1-\alpha)^2\sigma_M^2\right)^{1/2}.
\end{aligned}$$

The pair $(\mu_P, \sigma_P) = (\mu_P(\alpha), \sigma_P(\alpha))$ traces out a $\mu_P(\alpha)$ versus $\sigma_P(\alpha)$ curve in portfolio mean return versus standard deviation space as α varies from 0 to 1, with value (μ_i, σ_i) when $\alpha = 1$ and value (μ_M, σ_M) when $\alpha = 0$. This curve can never lie above or to the left of the CML (if it did, the CML would not be optimal), and the slope of the curve at $\alpha = 0$ is

$$\left.\frac{d\mu}{d\sigma}(\alpha)\right|_{\alpha=0} = \frac{d\mu/d\alpha|_{\alpha=0}}{d\sigma/d\alpha|_{\alpha=0}}, \tag{D.10}$$

where the derivative of the numerator is equal to $\mu_i - \mu_M$ for all α including $\alpha = 0$. Straightforward calculus yields an expression for $d\sigma/d\alpha$ that reduces to $\left(\sigma_{i,M} - \sigma_M^2\right)/\sigma_M$ when $\alpha = 0$. Thus

$$\left.\frac{d\mu}{d\sigma}(\alpha)\right|_{\alpha=0} = \sigma_M \frac{\mu_i - \mu_M}{\sigma_{i,M} - \sigma_M^2} \tag{D.11}$$

A small but important step in the proof is a straightforward verification that the derivative function $\frac{d\mu}{d\sigma}(\alpha)$ is continuous at $\alpha = 0$. Thus at $\alpha = 0$ the derivative $\frac{d\mu}{d\sigma}(\alpha)$ is tangent to the CML, and so its slope must equal that of the CML, or $(\mu_M - r_f)/\sigma_M)$. [1] It follows that

$$\mu_M - r_f = \sigma_M^2 \frac{\mu_i - \mu_M}{\sigma_{i,M} - \sigma_M^2}, \tag{D.12}$$

which, upon rearrangement, yields the CAPM.

D.4 Zero Beta CAPM Derivation

Here we provide a simple derivation of the Zero Beta CAPM using MinVar portfolio results in Section D.1. The assumptions we use are the same as those for the standard CAPM, except for the assumption that there is no risk–free asset available. A further difference from the standard CAPM setup is that different investors will take different positions on the efficient frontier. However, in equilibrium, the market portfolio M is a convex combination of efficient portfolios of all investors, and is therefore efficient.

[1] The reason is that if the curve of portfolio P is not tangent to the CML, its variance would be lower than the CML variance for μ_P in some neighborhood of μ_M, contradicting the optimality of the CML.

Recall from equation (D.1) that the weight vector of a MinVar efficient portfolio p, with return r_p and mean return μ_p, has the form

$$\mathbf{w}_p = \lambda \mathbf{C}^{-1} + \gamma \mathbf{C}^{-1}\mathbf{1}.$$

It follows that for any other portfolio q with weight vector $\mathbf{w}_q$, return r_q and mean return μ_q, the covariance of r_p and r_q is

$$\begin{aligned} \text{cov}(r_p, r_q) &= \mathbf{w}_q' \mathbf{C} \mathbf{w}_p \\ &= \lambda \mathbf{w}_q' + \gamma \mathbf{w}_q' \mathbf{1} \\ &= \lambda \mu_q + \gamma. \end{aligned} \tag{D.13}$$

A portfolio that is constrained to have zero covariance with respect to an efficient frontier portfolio p is called a *zero covariance portfolio*. A fully-invested zero-covariance portfolio $zc(p)$, that has weight vector $\mathbf{w}_{zc(p)}$, return $r_{zc(p)}$, and mean return $\mu_{zc(p)}$, satisfies

$$\begin{aligned} \text{cov}(r_p, r_{zc(p)}) &= \lambda \mathbf{w}_{zc(p)}' + \gamma \mathbf{w}_{zc(p)}' \mathbf{1} \\ &= \lambda \mu_{zc(p)} + \gamma \\ &= 0. \end{aligned} \tag{D.14}$$

A formula for a zero beta portfolio weight vector is derived in TS D.5, where the frontier geometray of zero Beta portfolios is discussed. Solving for γ in (D.14) and substituting in (D.13) gives

$$\text{cov}(r_p, r_q) = \lambda(\mu_q - \mu_{zc(p)}) \tag{D.15}$$

$$\text{var}(r_p) = \lambda(\mu_p - \mu_{zc(p)}). \tag{D.16}$$

Dividing (D.15) by (D.16), rearranging and let p be the market portfolio M, we have:

$$\mu_q = \mu_{zc(M)} + \beta_{q,M}(\mu_M - \mu_{zc(M)}). \tag{D.17}$$

Since the portfolio q is an arbitrary fully-invested portfolio, it can be an individual asset with index j , which results in the asset level zero-beta CAPM:

$$\mu_j = \mu_{zc(M)} + \beta_{j,M}(\mu_M - \mu_{zc(M)}). \tag{D.18}$$

D.5 Zero Covariance Frontier Portfolios

Any portfolio p that has zero covariance with respect to another portfolio q is a portfolio whose beta with respect to q is zero, i.e., $\beta_{pq} = 0$, and the portfolio p is called a *zero beta* portfolio. It turns out that a portfolio or asset that has a zero beta with respect to the market portfolio is the basis of a *zero beta CAPM*, which we discussed in the previous Section. Here we discuss the special case of zero covariance frontier portfolios. It turns out the zero covariance frontier portfolios are inefficient, and since the MaxQU approach only gives efficient frontier portfolios, we use instead the MinVar optimization results.

Let p denote any efficient frontier portfolio not equal to the global minimum variance portfolio, and define $zc(p)$ to be a frontier portfolio that has a zero covariance with respect to p. The expected value of the $zc(p)$ portfolio is easily found from general expression for the covariance between any two frontier portfolios, which in this case is

$$\begin{aligned}\operatorname{cov}(r_{zc(p)}, r_p) &= \sigma^2_{\text{gmv}} + \frac{c}{d}\left(\mu_{zc(p)} - \mu_{\text{gmv}}\right)\left(\mu_p - \mu_{\text{gmv}}\right) \\ &= 0\end{aligned}$$

where $c = \mathbf{1}'\mathbf{C}^{-1}\mathbf{1} > 0$ and the formula for the positive constant d is given by (D.5). It follows that the expected return of the zero covariance frontier portfolio is given by the formula:

$$\mu_{zc(p)} = \mu_{\text{gmv}} - \frac{d}{c} \cdot \frac{\sigma^2_{\text{gmv}}}{\mu_p - \mu_{\text{gmv}}}. \tag{D.19}$$

Since p is any efficient frontier portfolio other than the GMV portfolio it follows that $\mu_p > \mu_{\text{gmv}}$. Thus the second term above is positive and $\mu_{zc(p)} < \mu_{\text{gmv}}$, i.e., the mean return of the zero covariance frontier portfolio is always less than that of the GMV portfolio. This means that the zero covariance portfolio is an inefficient frontier portfolio, i.e., it lies on the lower half of the portfolio frontier that corresponds to the minus sign in . The zero covariance portfolio weight vector $\mathbf{w}_{zc(p)}$ corresponding to $\mu_{zc(p)}$ is uniquely defined by (D.3) as:

$$\mathbf{w}_{zc(p)} = \mathbf{w}_1 + \mu_{zc(p)}\mathbf{w}_2. \tag{D.20}$$

It turns out that the relationship between the efficient frontier portfolio p, the GMV portfolio p_{gmv} and the zero covariance portfolio $zc(p)$ is described by the straight line in Figure D.1. Note in particular that the straight line connecting portfolios $zc(p)$ and p goes through the portfolio p_{gmv}. NOTE: In the figure the *gmv* is referred to as the MVP.

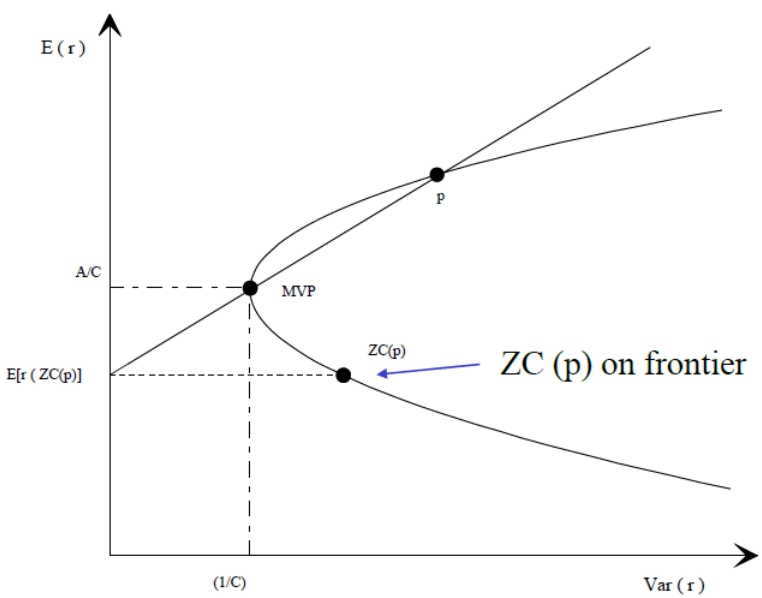

Fig. D.1 Relationship between portfolios p, p_{gmv} and $zc(p)$ in mean return versus variance coordinates.

It can be shown that the straight line in the above figure is given by

$$\mu_r = \mu_{zc(p)} + \frac{\mu_p - \mu_{gmv}}{\sigma_p^2 - \sigma_{gmv}^2}\sigma_r^2 \tag{D.21}$$

and that $\mu_r = \mu_{gmv}$ when $\sigma_r^2 = \sigma_{gmv}^2$.

D.6 Proof of the APT

The vector $\mathbf{a} = (a_1, a_2, \cdots, a_n)'$ of intercepts in may be written in the least-squares regression fit-plus-residuals form

$$\mathbf{a} = \hat{\lambda}_0\mathbf{1} + \mathbf{B}\hat{\boldsymbol{\lambda}} + \mathbf{u} \tag{D.22}$$

where $\hat{\lambda}_0$ and $\hat{\boldsymbol{\lambda}} = (\hat{\lambda}_1, \hat{\lambda}_2, \cdots, \hat{\lambda}_K)'$ are the least-squares regression coefficients, and $\mathbf{u} = (u_1, u_2, \cdots, u_n)'$ is the vector of regression residuals. For details on the above LS regression decomposition see , where it is shown that the regression residuals vector $\mathbf{u}$ is orthogonal to the vector $\mathbf{1}$ as well as to the columns of $\mathbf{B}$. Thus $\mathbf{u}'\mathbf{1} = \sum_{i=1}^n u_i = 0$, and $\mathbf{u}'\mathbf{B} = \mathbf{0}'$ where the right hand side is a vector of K zeros. It follows of course that $\mathbf{u}$ is orthogonal to the fit $\mathbf{a} = \hat{\lambda}_0\mathbf{1} + \mathbf{B}\hat{\boldsymbol{\lambda}}$.

Consider the zero investment portfolio weights vector

$$\mathbf{w} = \frac{\mathbf{u}}{\sqrt{n}\|\mathbf{u}\|} \tag{D.23}$$

where $\|\mathbf{u}\| = \sqrt{\sum_{i=1}^n u_i^2}$. Then using along with the fact that $\mathbf{u}'\mathbf{B} = \mathbf{0}'$, the arbitrage portfolio p has return

$$\begin{aligned} r_p &= \mathbf{w}'\mathbf{r} \\ &= \frac{1}{\sqrt{n}\,\|\mathbf{u}\|}(\mathbf{u}'\mathbf{a} + \mathbf{u}') \end{aligned} \tag{D.24}$$

and since $\mathrm{E}() = 0$, the arbitrage portfolio has mean

$$\begin{aligned} \mu_p &= \frac{\mathbf{u}'\mathbf{a}}{\sqrt{n}\,\|\mathbf{u}\|} \\ &= \frac{\mathbf{u}'\mathbf{u}}{\sqrt{n}\,\|\mathbf{u}\|} \\ &= \frac{1}{\sqrt{n}}\,\|\mathbf{u}\| \end{aligned} \tag{D.25}$$

where the second line is due to substituting (D.22) for $\mathbf{a}$ and using the orthogonality of $\mathbf{u}$ to $\mathbf{1}$ and $\mathbf{B}$, and the last line is because $\mathbf{u}'\mathbf{u} = \|\mathbf{u}\|^2$.

The variance of the arbitrage portfolio r_p is

$$\begin{aligned} \mathrm{E}(r_p - \mu_p)^2 &= \mathrm{E}\left(\frac{\mathbf{u}'}{\sqrt{n}\,\|\mathbf{u}\|}\right)^2 \\ &= \frac{\mathbf{u}'\mathbf{C}\mathbf{u}}{n\,\|\mathbf{u}\|^2} \end{aligned} \tag{D.26}$$

where $\mathbf{C}$ is the covariance matrix of the errors $= (\varepsilon_1, \varepsilon_2, \cdots, \varepsilon_n)'$. We say that the arbitrage portfolio is *asymptotically risk diversifiable* (ARD) if

$$\lim_{n\to\infty} \frac{\mathbf{u}'\mathbf{C}\mathbf{u}}{n\,\|\mathbf{u}\|^2} = 0. \tag{D.27}$$

The simplest condition that results in ARD is where the ε_i are uncorrelated and have bounded variances $\sigma_i^2 < B$, for in that case $\mathbf{u}'\mathbf{C}\mathbf{u} = \sum_{i=1}^n u_i^2\sigma_i^2 < B\sum_{i=1}^n u_i^2$ and the above ratio is bounded by B/n, which goes to 0 as $n \to \infty$.

Suppose the portfolio is ARD and that

$$\begin{aligned}
\mu_{p,\infty} &= \lim_{n\to\infty} \frac{1}{\sqrt{n}}\,\|\mathbf{u}\| \\
&= \lim_{n\to\infty} \left(\frac{1}{n}\sum_{i=1}^n u_i^2\right) \\
&> 0.
\end{aligned}$$

In that case, the arbitrage portfolio has an asymptotic sure profit with no risk. But in a market that is arbitrage free this is not possible, and it must be that the APT condition

$$lim_{n\to\infty}\left(\frac{1}{n}\sum_{i=1}^n u_i^2\right) = 0$$

is satisfied.

D.7 Robust Estimation Details

D.7.1 mOpt Psi Function Formula

The mOpt psi function formula is

$$\psi_{mOpt}(x) = \begin{cases} x & |x| \leqslant 1 \\ \dfrac{\varphi(1)}{\varphi(1)-a}\left(x - \mathrm{SGN}(x)\dfrac{a}{\varphi(x)}\right)U(c-|x|) & |x| > 1 \end{cases} \tag{D.28}$$

where $\varphi(x)$ is the standard normal density function, $U(x)$ is the unit step function whose value is 1 for $x \geqslant 0$ and is 0 for $x < 0$, and the constants a and c depend on the desired normal distribution efficiency. Note that $\psi_{mOpt}(x) = 0$ for x outside the interval $(-c, c)$. A 95% normal distribution efficiency is obtained with $a = 0.0132$ and $c = 3.004$. The rho function $\rho_{mOpt}(x)$ is obtained by an integration of $\psi_{mOpt}(x)$, and because the latter is zero outside the interval $(-c, c)$, the rho function is bounded with a constant value outside that interval. Details of the integral of the mOpt psi function, and computing values of a and c from specified normal distribution efficiencies are provided in **KonisMartin2021**.